Health-a-thon 2026

FINAL SUBMISSION

GlucoFlow

Gluco: a Self Evolving healthcare operations agent

TEAM NAME BetaRevive

TEAM CODE 7BUKY8

Team Overview & Background

- Rai Biswas | Team leader and ideation
- Led the core ideation and shaped the direction of GlucoFlow.
- Dr. Manikanda Krishnan | Doctor partner
- Helped validate the idea and provided a clinical perspective during concept development.
- Vishruth (wysh) | Technical lead and builder
- Defined the technical scope, built the application and drove implementation improvements.
- Together, the team developed a shared care workspace around the proposed coordination problem. Doctor input supported idea validation; no clinical trial or hospital outcome validation is claimed.

Problem Statement and Supporting Evidence

- The coordination gap: a recorded order does not establish that the next step happened.
- A specimen may be collected but not received by the lab. A validated result may still await acknowledgement.
- At shift change, staff must reconstruct the evidence and identify who owns the next step.
- DKA/HHS provides a time-sensitive context for these operational handoffs. Clinical interpretation stays with the treating team.
- Our target: reduce time spent chasing unresolved handoffs, measured against the ward's baseline.

Target Users and Use Case

- Doctor: inspect assigned encounters, approve shift-wide monitoring and edit handovers.
- Nurse: see assigned exceptions, record collection and acknowledge awareness without falsely closing work.
- Laboratory: track expected and received specimens, then record result validation.
- Coordinator: track receiving responsibility and bed/transfer status separately.
- Administrator: manage staff access. Each action retains the acting person and role.
- Demo setting: fictional diabetes-related inpatient encounters with simulated hospital and lab feeds.

Solution Overview, Key Features & Impact

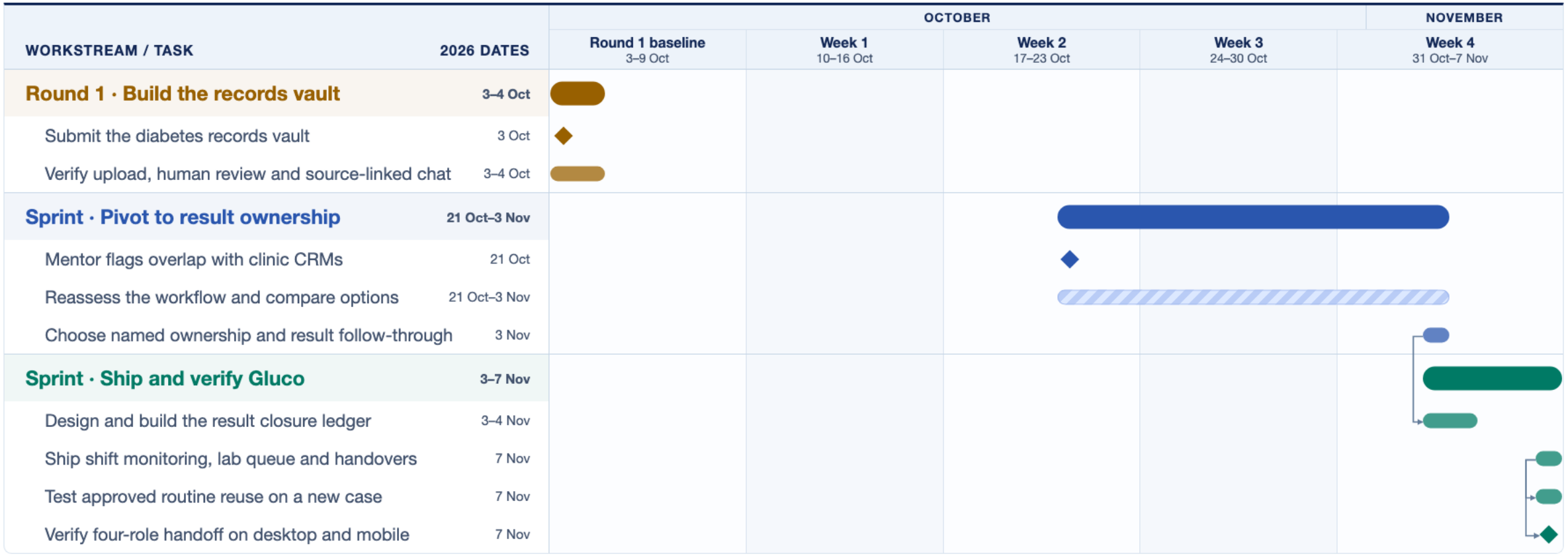
- Ask: ‘Monitor unacknowledged results for my assigned patients until shift end.’ Inspect the scope and approve one task.
- See the missing step, responsible team and source in the care pipeline. A matching event confirms completion.
- Capture a confirmed outcome as a Care Routine. Test it, approve a version and reuse it for another encounter.
- Reuse requires that encounter’s own evidence and a fresh approval. Versions support rollback.
- Differentiators: evidence-based closure, visible responsibility across teams and governed workflow learning.

Technology, Tools and AI Approach

- Gluco is a Self Evolving healthcare operations agent built on a customized Hermes runtime.
- Its learning loop captures confirmed outcomes as Care Routines, then supports testing, approval, reuse and rollback.
- Intelligent UI brings encounter cards, source evidence and workflow actions into the conversation.
- The backend checks authority and evidence before actions. PostgreSQL retains approvals and audit history.
- Dots and Codex inspire the workspace experience. Today, self-evolution is bounded procedure reuse. Broader staff-driven adaptation is the roadmap.

React / TypeScript • OpenUI • Fastify • PostgreSQL • Sarvam speech. Synthetic HIS/LIS feeds.

Build Sprint Journey and Progress



— **Documented span** ◆ **Milestone** ↳ **Delivery sequence** Phase bars span documented checkpoints. The striped bar shows the review window between dated notes.

How Gluco evolves

Capture a routine → Test it → Clinician approves → Reuse on a new case

Approved specimen and result routines are versioned. Each new case needs its own evidence and approval.

Evaluation and User Validation

- 420 automated tests passed across 78 files on 7 November.
- 204 distinct browser cases passed across runs. Two desktop-only cases were skipped on mobile.
- Separate authenticated roles exercised collection, lab receipt, result validation and acknowledgement.
- Hosted checks passed monitoring edits/stops, exact-version handover receipt and approved routine reuse.
- Tests covered access isolation, duplicate events, stale revisions and corrected results reopening work.
- Engineering evidence uses fictional data. No real-world time savings or clinical outcomes are claimed.

Next validation: clinician usability, live speech/model reliability, measured latency and operating cost.

Safety and Responsible AI

- Assistive operational scope: no diagnosis, prescribing, clinical risk scores or treatment decisions.
- The public demo uses fictional records and shared demo accounts. No live ABHA/ABDM connection.
- Server-side role and encounter checks enforce authority. Human approval gates operational actions.
- Viewing, staff acknowledgement and source-confirmed completion remain distinct states.
- Missing or stale evidence stays unverified. Corrections retain history and can reopen pending work.
- Routine learning cannot rewrite clinical policy. Hospital privacy review and language validation precede real use.

Deployment, Cost and Sustainability Plan

- Deployment today: web on Vercel, API and workers on Azure, PostgreSQL/Supabase services. HIS/LIS feeds are simulated.
- Pilot entry: one ward with named owners, agreed source events and a privacy/security review.
- Planning assumption: ₹40,000 setup, based on 40 hours at ₹1,000/hour. Connector development is additional.
- Illustrative monthly cost: ₹6,000/facility, comprising ₹3,000 hosting, ₹2,000 AI/speech and ₹1,000 support.
- Proposed revenue: a facility subscription. Validate usage cost, staff effort and willingness to pay before pricing.

Next Steps

- Days 1–30: agree handoff ownership, validate event timestamps and measure the existing workflow.
- Days 31–60: run read-only shadow mode, review exceptions and test proposed Care Routines.
- Days 61–90: conduct a limited supervised pilot and compare delays with baseline. Track chasing time, missed handoffs and alert burden.
- Learning roadmap: staff corrections become proposed routine changes, with unit-owner approval and post-release monitoring.
- Expand to discharge and referral continuity only after source reliability and usability meet agreed criteria.

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Thank you

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https://glucoflow.vercel.app/
Select Explore the clinic. No registration or credentials required.
Fictional data only. Hospital and lab updates are simulated.
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